

Duke University

The Geometry of Reasoning

Flowing Logics in Representation Space

Yufa Zhou*, Yixiao Wang*, Xunjian Yin*, Shuyan Zhou, Anru R. Zhang

*Equal contribution

ICLR 2026



Roadmap

- | | | |
|----|---------------------|--|
| 01 | Motivation | Reasoning now does research mathematics — and we still cannot see how. Prior interpretations are discrete, local, or static. |
| 02 | Background | Concept spaces, linear representations, and representation manifolds: geometry of meaning in LLMs. |
| 03 | Theory | Reasoning as a smooth flow on a manifold; logic as differential constraints on that flow. |
| 04 | Experiments | A carrier-invariant logic dataset; position / velocity / curvature across six LLMs. |
| 05 | Significance | Evidence against a purely surface account, and where this line of research goes next. |

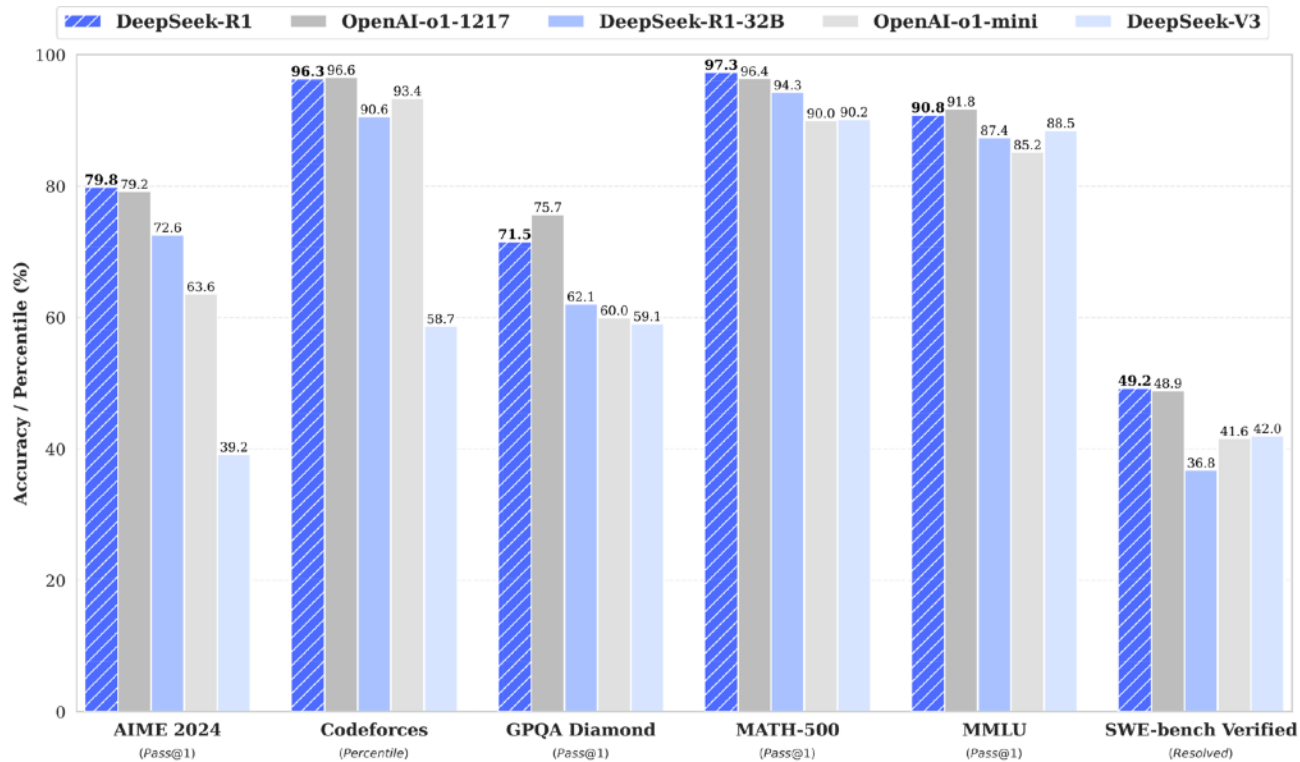
01

Motivation

Reasoning models are remarkably capable, yet we still lack a dynamic account of how reasoning unfolds inside the model. Topology tells us what is connected; it cannot tell us how far, how fast, or how sharply.

Capability · Opacity · Prior views · Core question

Reasoning models are extremely powerful



Reasoning models reach near-expert accuracy on competition math, code and science benchmarks.

“

We achieved gold medal level performance on the 2025 IMO competition with a general-purpose reasoning system.

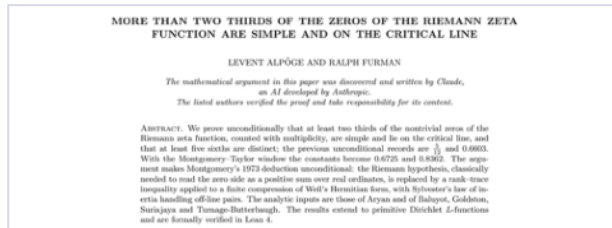
This is an LLM doing math, not a specific formal math system.

Sam Altman, July 2025

Gold-medal IMO performance from a **general-purpose** reasoning system — not a formal math solver.

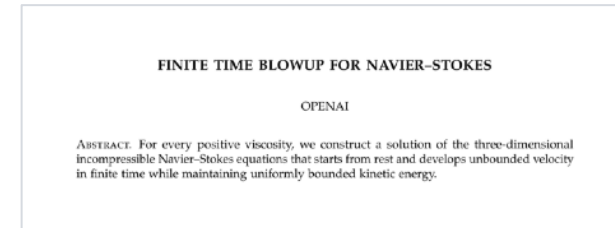
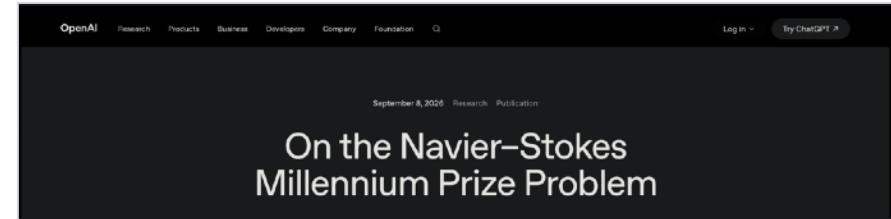
Sources: DeepSeek-R1 technical report; @sama, Jul 2025.

2026: machines are doing research mathematics



AUG 2026 · ANTHROPIC

- The argument was **discovered and written by Claude**; the listed authors verified it and take responsibility for its content.
- Simple zeros on the critical line: $5/12 \rightarrow 2/3$ (0.6725 with the Montgomery-Taylor window). Formally verified in **Lean 4**.

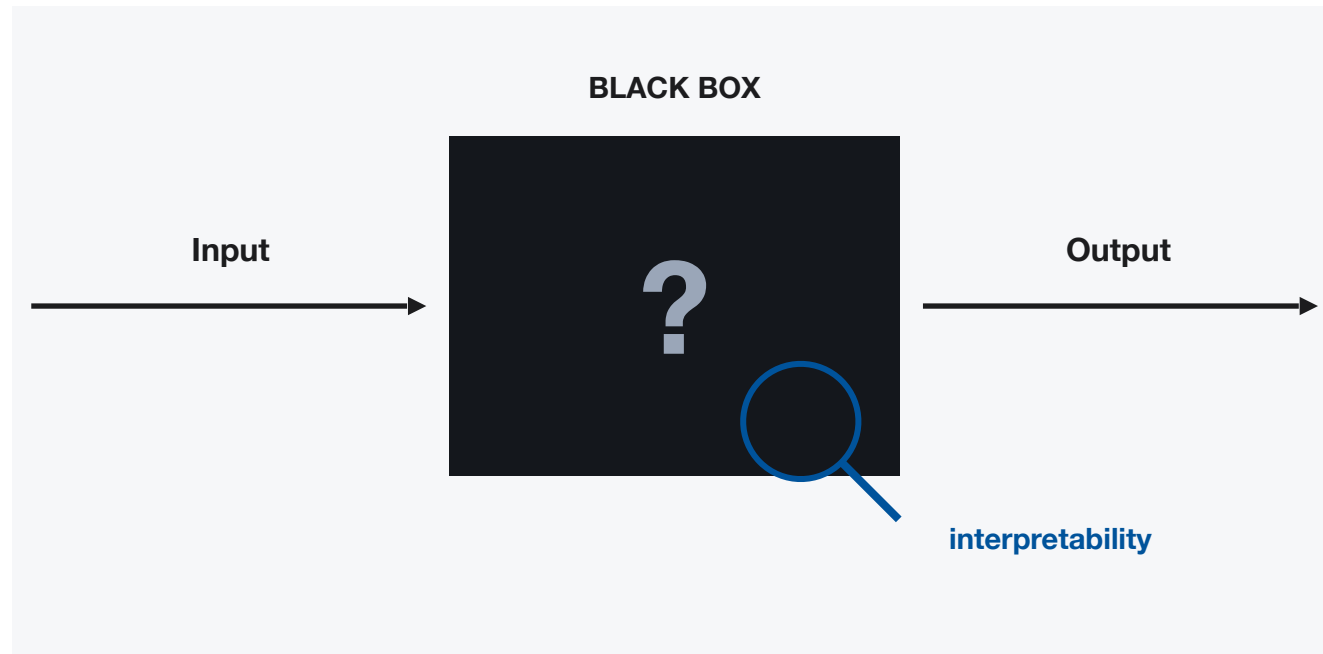


8 SEP 2026 · OPENAI

- Establishes **alternatives (C) and (D)** of Fefferman's statement — blow-up **under smooth forcing**; the unforced question **remains open**.
- Lean-formalised; concurrent Euler results by Buckmaster & Alpöge, with **credit still disputed**.

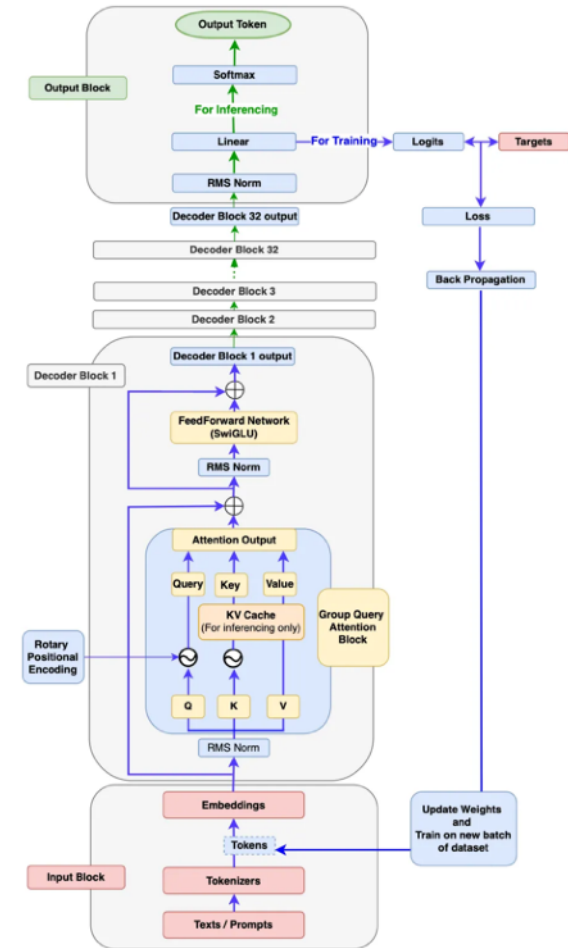
We can now **verify** what these systems produce. We still cannot **explain** how they produced it — and that gap is what this research is about.

Yet LLMs are still black boxes



THE GAP

We can measure **what** a model outputs, but not **why** a particular chain of thought was produced.



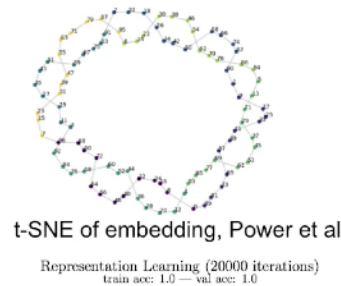
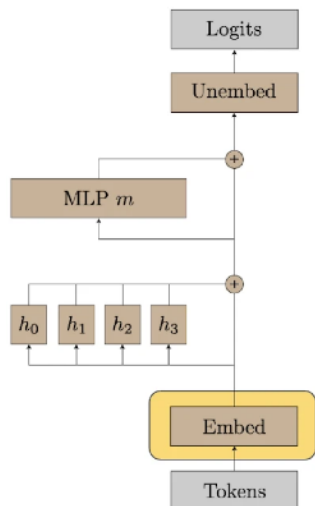
Every component is fully specified — and still not interpretable.

How can we analyze and understand LLMs?

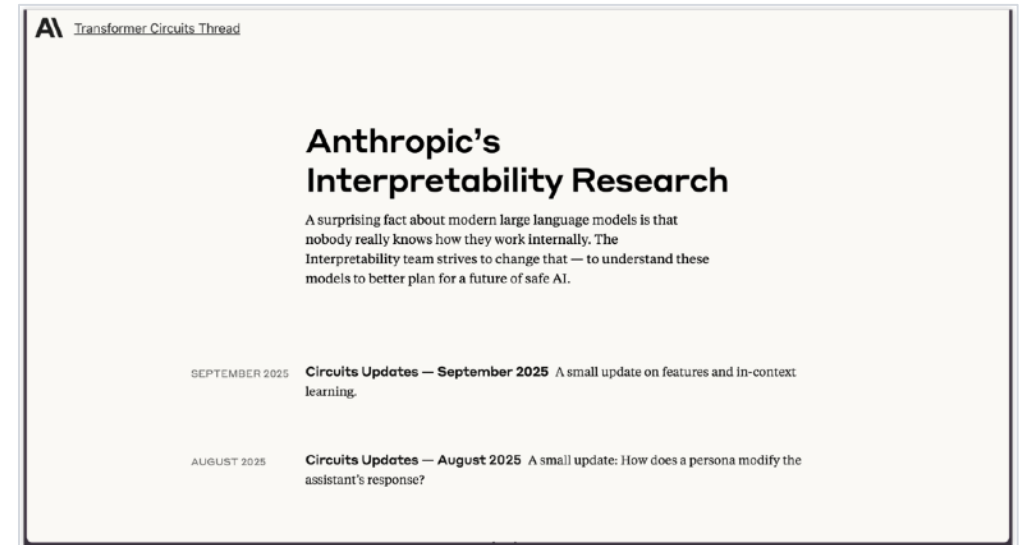
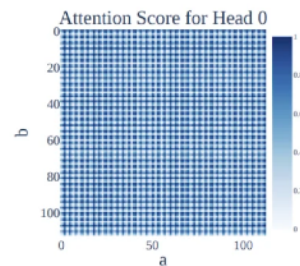
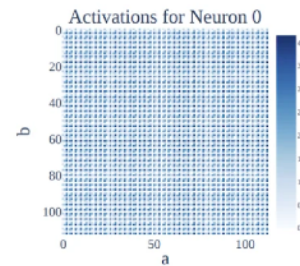
MECHANISTIC INTERPRETABILITY

Reverse-engineer the features a model has learned. Use microscopic tools — linear probes, sparse autoencoders, circuit analysis — and study AI as a **natural science**.

Understanding the Embedding



Principal components of embedding, Liu et al



A sustained research program (Anthropic, Transformer Circuits).

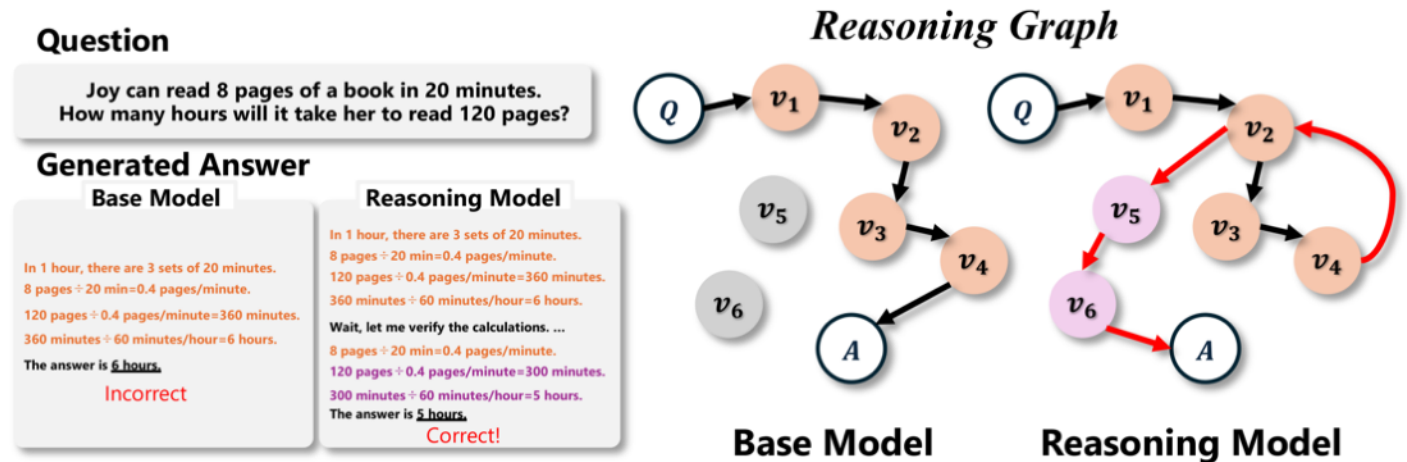
Probing embeddings, activations and attention to recover interpretable structure.

Previous interpretation of reasoning

Existing view

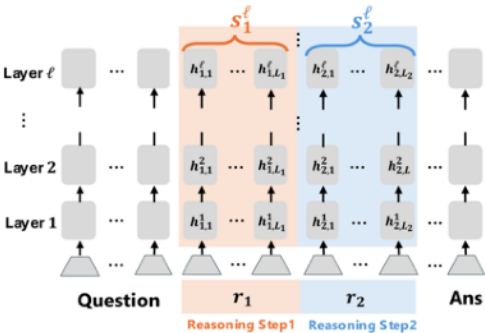
Reasoning is a random walk on a graph.

- Reasoning steps are clustered into **nodes**; transitions between them form a **reasoning graph**.
- Reasoning models produce graphs that are more **cyclic**, larger in **diameter**, and more **small-world**.
- These graph-theoretic properties correlate with performance on hard math tasks.



Minegishi, G., Furuta, H., Kojima, T., Iwasawa, Y., & Matsuo, Y. (2025). Topology of Reasoning: Understanding Large Reasoning Models through Reasoning Graph Properties. NeurIPS 2025.

How the reasoning graph is built — and what it misses

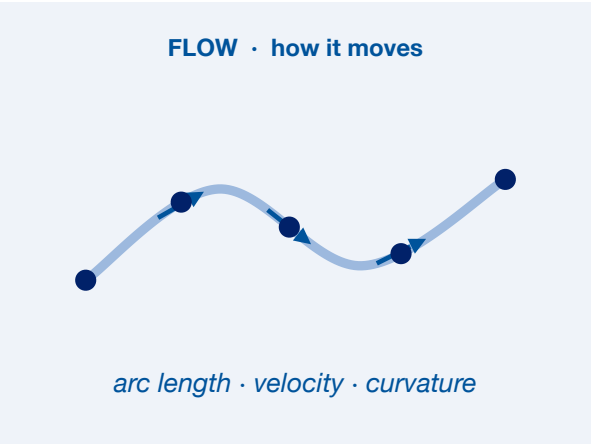
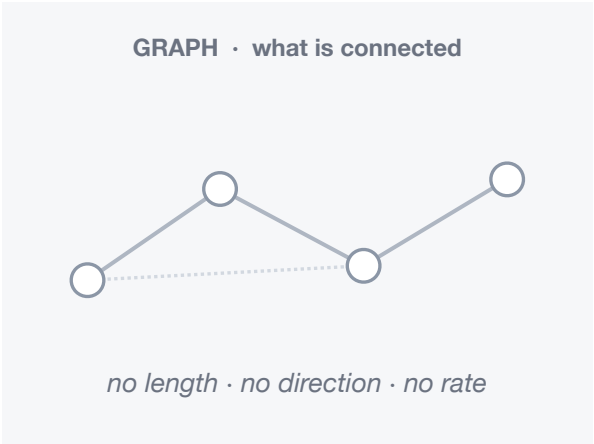


(a) Extract reasoning graphs from LLMs

Node	Generated Reasoning Step
Multiplicative (Node 4)	First leg: $10 \times 30 = 300$. Now multiply that by 120×1.157625 .
Additive (Node 21)	Total chairs: $22 + 66 = 88$. Total sofas: $20 + 64 = 84$.
Wait (Node 15)	Wait, but we need to add this ... Wait, no. Let me clarify ...

(b) Representative Nodes from Reasoning Graph

Segment hidden states are mean-pooled per reasoning step, then K-means clustered into graph nodes.



WHAT IS THROWN AWAY

Clustering discards **length, direction and rate of change**. A graph records which states are connected — never how the representation actually **moves** between them.

Topology has no metric. Geometry does.

Topology has no length — what if geometry?

LLM reasoning looks **discrete** (tokens), but language representations and embeddings are **continuous**.

“Consciousness does not appear to itself chopped up in bits. ... It is nothing jointed; it flows. A river or a stream are the metaphors by which it is most naturally described.”

William James, The Principles of Psychology (1890)

EXISTING VIEWS

Graphs

discrete, noisy

Circuits

local, microscopic,
messy

Embeddings

static

MISSING

A **dynamic** view explaining **how** reasoning evolves in representation space.

CORE QUESTION

Do LLMs internally treat reasoning as **continuous flows** — and what controls these flows?

02

Background

Before asking how reasoning moves, we need the space it moves in. Three decades of work — from conceptual spaces to the linear representation hypothesis — say that meaning already has geometry.

Concept spaces · Linear structure · Polytopes · Manifolds · Symmetry

Concept space algebra

What is a concept space (i.e. language representations)?

- Information is organised into **high dimensions** — as today's LLMs do.
- Concepts are sorted into **domains**: numbers, space, time, temperature, colour, nouns, ...
- Dimensions within a domain stand in **geometric relations** — parallel, orthogonal, ...
- Domains are endowed with a **topology or metric**.
- Similarity is represented by **distance** in conceptual space.

Concept-space theory has deep roots in cognitive psychology. Modern LLMs let us **test it empirically** for the first time.



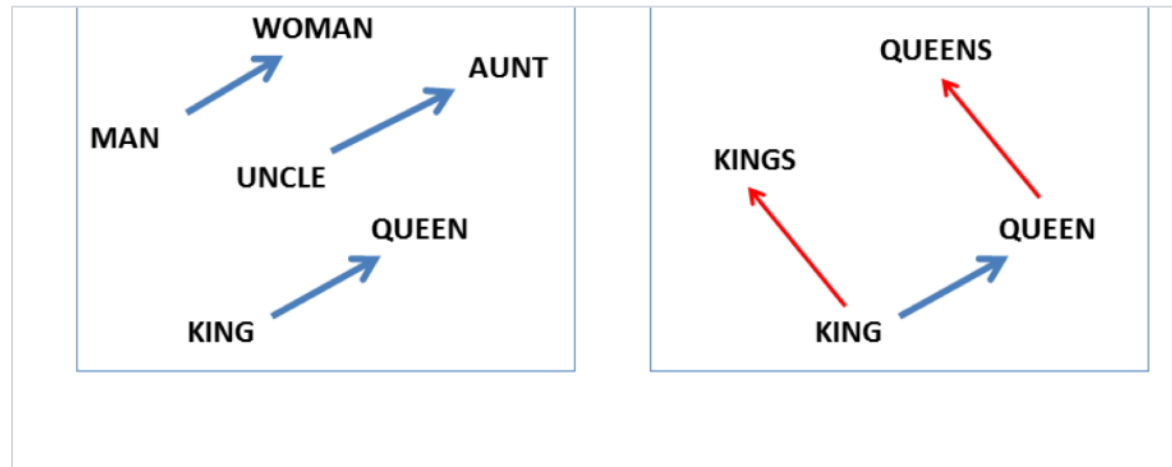
Gärdenfors (2014)



Gärdenfors (2000)

Linear algebra on concept space

king – queen $\approx$ man – woman



Efficient Estimation of Word Representations in Vector Space

Tomas Mikolov
Google Inc., Mountain View, CA
tmikolov@google.com

Kai Chen
Google Inc., Mountain View, CA
kaichen@google.com

Greg Corrado
Google Inc., Mountain View, CA
gcorrado@google.com

Jeffrey Dean
Google Inc., Mountain View, CA
jeff@google.com

Linguistic Regularities in Continuous Space Word Representations

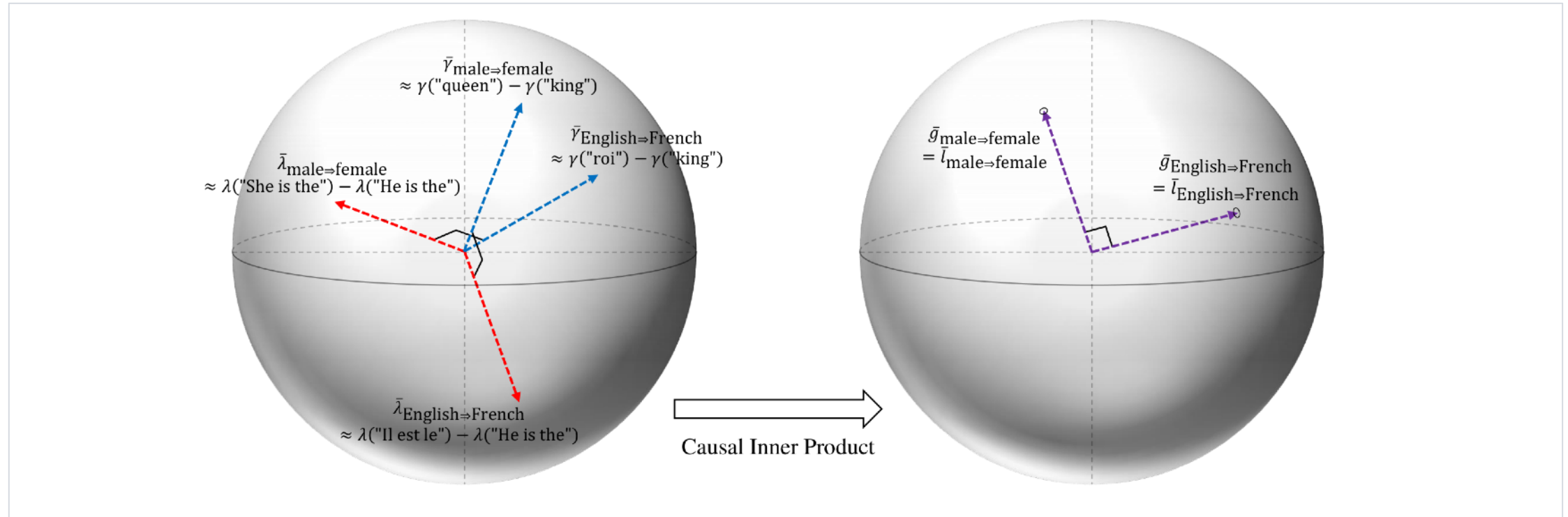
Tomas Mikolov*, Wen-tau Yih, Geoffrey Zweig
Microsoft Research
Redmond, WA 98052

Linear Algebraic Structure of Word Senses, with Applications to Polysemy

Sanjeev Arora, Yuanzhi Li, Yingyu Liang, Tengyu Ma, Andrej Risteski
Computer Science Department, Princeton University
35 Olden St, Princeton, NJ 08540
{arora, yuanzhil, yingyul, tengyu, risteski}@cs.princeton.edu

Semantic relations appear as **vector offsets** — meaning is already structured linearly in embedding space.

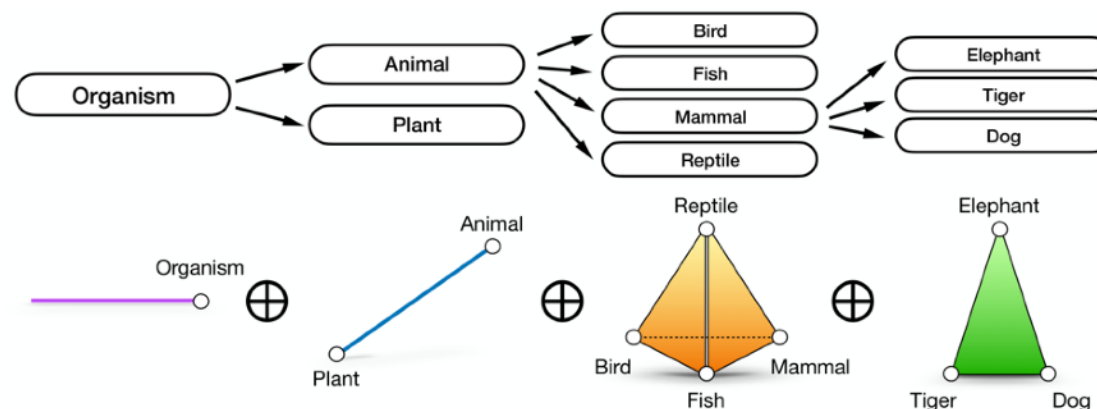
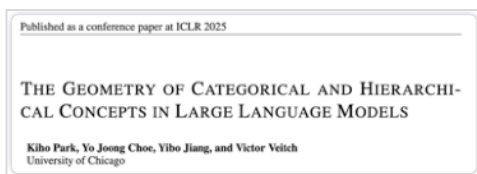
The linear representation hypothesis



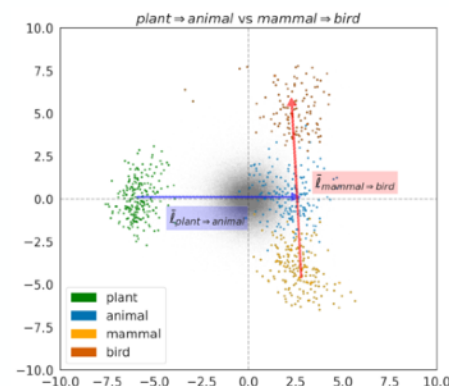
Concepts have linear representations in both embedding and unembedding space; under a **causal inner product**, causally separable concepts become **orthogonal** vectors.

From binary directions to categorical polytopes

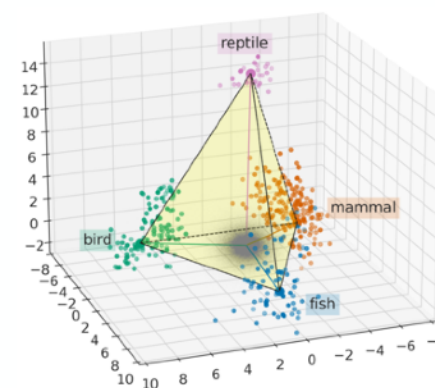
- Prior work formalised **binary concepts** with natural contrasts — e.g. {male, female} — as **directions** in representation space.
- Yet many concepts admit no such contrast (e.g. “is animal”).
- This work extends the theory to encode features as **vectors**, so that **categorical concepts** are represented as **polytopes**.
- Hierarchy becomes **orthogonality**: plant $\Rightarrow$ animal and mammal $\Rightarrow$ bird live in orthogonal subspaces.



(a) Pictorial depiction of the representation of hierarchically related concepts.



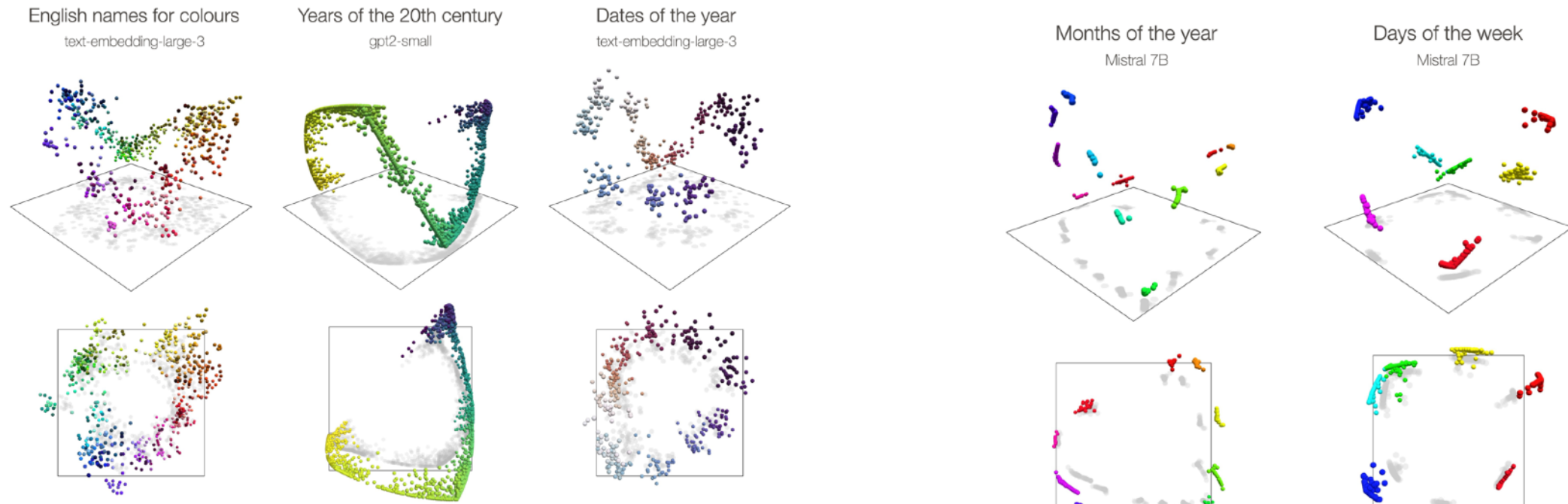
(b) Hierarchy is encoded as orthogonality in Gemma. (c) Categorical concepts are represented as polytopes in Gemma.



Park, K., Choe, Y. J., Jiang, Y., & Veitch, V. (2025). The Geometry of Categorical and Hierarchical Concepts in Large Language Models. ICLR 2025.

The origins of representation manifolds

Embeddings in LLMs exhibit **manifold** structure strikingly aligned with human concept spaces — colours, years, months, dates.



Modell, A., Rubin-Delanchy, P., & Whiteley, N. (2025). *The Origins of Representation Manifolds in Large Language Models*. *arXiv:2505.18235*.

Why do these manifolds exist? Symmetry in language statistics

Symmetry in language statistics shapes the geometry of model representations

Dhruva Karkada^{1*} Daniel J. Korchinski² Andres Nava³ Matthieu Wyart^{2,3} Yasaman Bahri⁴

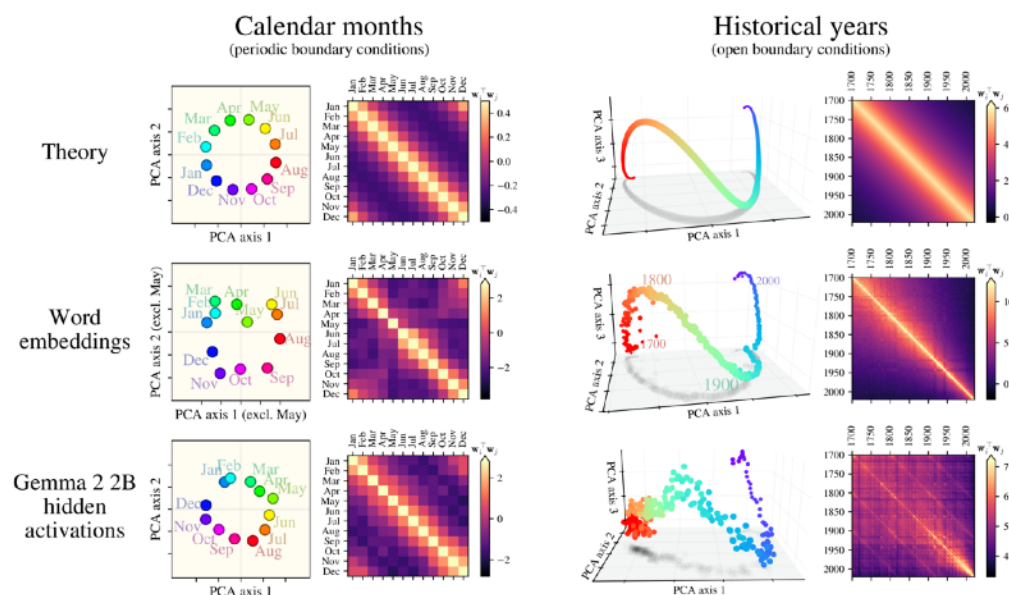


Figure 1: predicted by theory (top), found in word embeddings (middle) and in Gemma 2 2B (bottom).

Karkada, D., Korchinski, D. J., Nava, A., Wyart, M., & Bahri, Y. (2026). Symmetry in Language Statistics Shapes the Geometry of Model Representations. ICML 2026 (arXiv:2602.15029).

- Co-occurrence statistics carry a **translation symmetry** — how often two months co-occur depends only on the **interval** between them.
- They **prove** this symmetry generates the manifold, and **derive its geometry analytically**.
- It **survives** deleting every month-month co-occurrence — a **latent variable** is doing the work.

THE QUESTION IT RAISES

So representation geometry is real, and it is rich — at the level of **words**.

Does **reasoning** carry a structure of the same kind?

How do we study concept-space geometry for reasoning?

PRIOR WORK

Individual words and phrases

- Microscopic view
- Static, token-level geometry



OUR FOCUS

Reasoning at the sentence level

- Macroscopic behaviour
- A higher-level, abstract cognitive process

REQUIREMENT

We therefore need a **principled method to extract embeddings or hidden states that correspond to reasoning (chain-of-thought) processes.**

03

Theory

We model a chain of thought as a curve in representation space: discrete steps sampled from an underlying smooth trajectory, whose derivatives are constrained by logic.

Reasoning as flows · Formal hypothesis · Logic as constraints · Testable predictions

Reasoning as flows: extracting the trajectory

PROMPT

“The product of a set of distinct positive integers is 84. What is the least possible sum of these integers?”

ANSWER

Step 1: Factor $84 = 2 \times 2 \times 3 \times 7$. y_1

Step 2: We need distinct positive integers; duplicate factors like two 2's are not allowed as separate numbers. y_2

Step 3: To keep numbers distinct, we can merge prime factors: e.g. combine the two 2's as 4. y_3

Step 4: Candidate set $S = \{4, 3, 7\}$ because $4 \times 3 \times 7 = 84$ and all are distinct. y_4

- Each reasoning step is a **sentence**; we take its **mean-pooled** hidden state.
- The context **accumulates**: y_t encodes the whole prefix, not the step alone.

Algorithm 1: Get Context Cumulative Reasoning Trajectory

$\mathcal{V}$: vocabulary space; $P \in \mathcal{V}^n$: tokenized problem prompt; T : number of reasoning steps; $x_t \in \mathcal{V}^*$: tokens for step t ; $\mathcal{E} : \mathcal{V}^* \rightarrow \mathbb{R}^d$: representation operator; $y_t \in \mathbb{R}^d$: embedding at step t .

Input: $P \in \mathcal{V}^n$; $X = [x_1, \dots, x_T]$ with $x_t \in \mathcal{V}^*$

Output: $Y = [y_1, \dots, y_T] \in \mathbb{R}^{d \times T}$

$Y \leftarrow []$, $S_0 \leftarrow [P]$;

for $t \leftarrow 1$ **to** T **do**

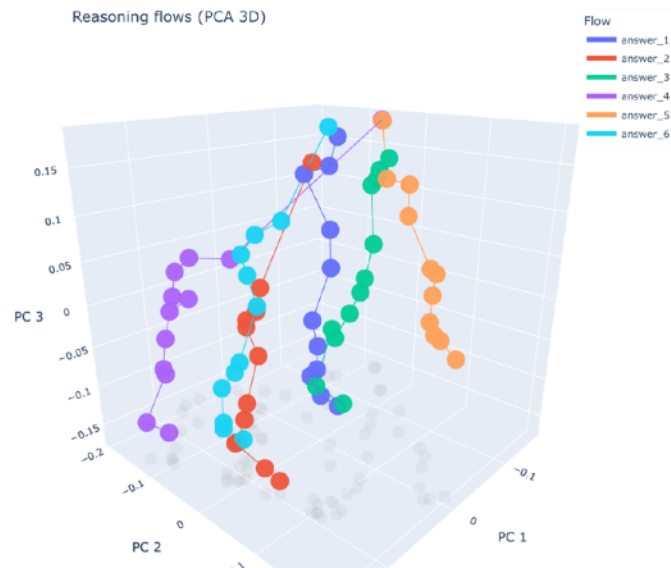
$S_t \leftarrow \text{Concat}(S_{t-1}, x_t)$; // Concatenate with previous context

$y_t \leftarrow \mathcal{E}(S_t)$; // Get embedding of current step

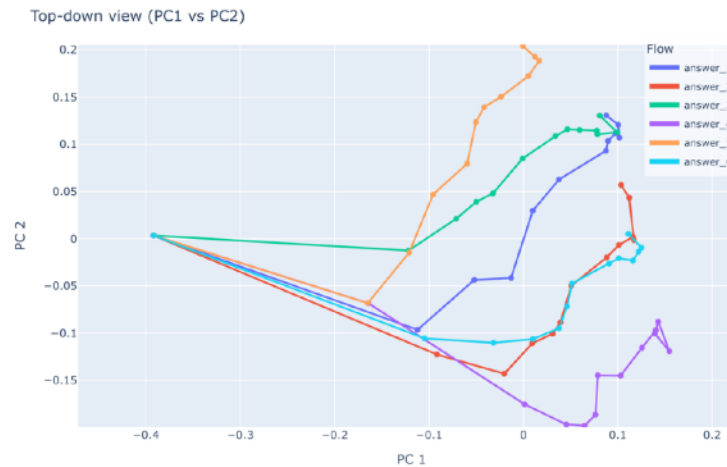
 Append y_t to Y ;

return Y ;

The extracted trajectories look like flows



(a) Reasoning flows visualized using PCA in 3 dimensions.



(b) Reasoning flows visualized using PCA in 2 dimensions.

- Six answers to the same prompt, projected by **PCA**.
- Trajectories are **smooth and directed** — not random jumps between clusters.
- Different answers trace **different but coherent** paths.

This is the empirical hint that motivates a **C¹ smooth-trajectory** model.

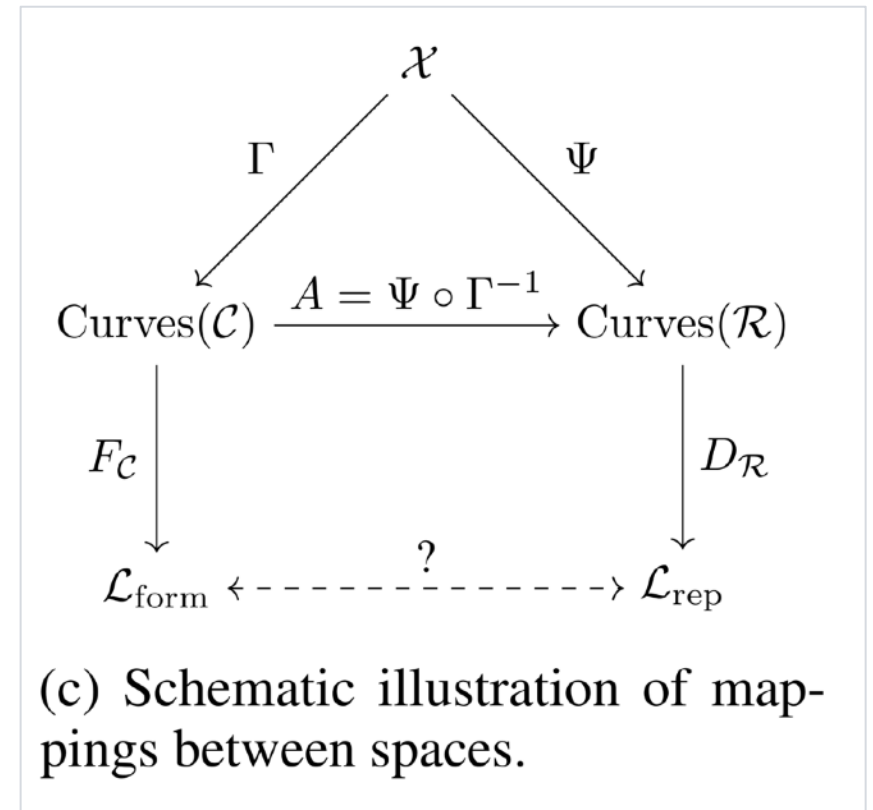
Formal hypothesis

Representations → smooth trajectories

- We define **context-cumulative embeddings** y_t that record the model's hidden state as the reasoning prefix grows.
- **Hypothesis:** these discrete points lie on an underlying **C¹ smooth trajectory** in representation space.

Concept space & flow

- **Input space** $\mathcal{X}$: natural-language sentences.
- **Concept space** $\mathcal{C}$: abstract semantic manifold capturing evolving reasoning states.
- **Representation space** $\mathcal{R}$: LLM embedding manifold obtained from hidden states.
- **Logic space** $\mathcal{L}$: structure of formal reasoning, living in the *differential geometry* of both $\mathcal{C}$ and $\mathcal{R}$; it constrains how flows evolve.



Logic as differential constraints on flow

Logic is **not an external module**. It governs how representations evolve step by step — we model it as **differential constraints on semantic flow**.

What the constraint couples

- **Discrete** reasoning structure, from humans
- **Continuous** geometric dynamics, in LLM representations

Why — the Wittgenstein reading

- “The world is the totality of facts, not of things” (*Tractatus*)
- Logical form is the **substrate** of language and reality

THE CLAIM UNDER TEST

In this spirit, logic becomes a **carrier-invariant skeleton** of reasoning. We test whether LLMs, trained on large corpora, internalise such structural invariants on the embedding manifold — effectively **rediscovering universal logical structure from data**.

Differential-geometric metrics

0

Position (0-order)

The original high-dimensional embedding vector y_t .

1

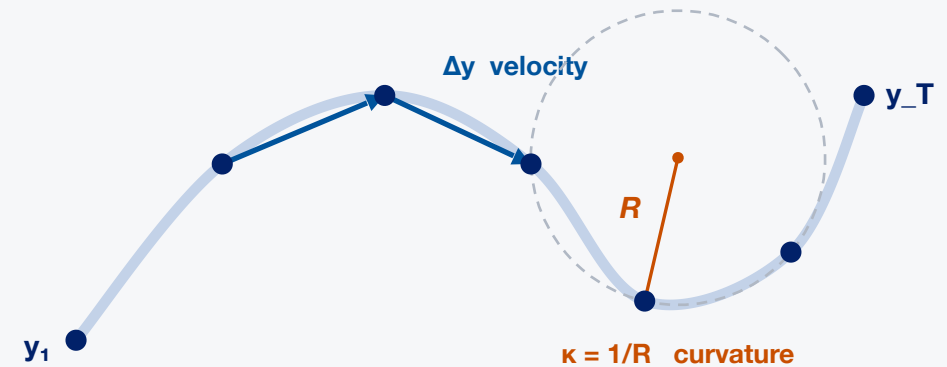
Velocity (1-order)

The difference between consecutive vectors, $y_{t+1} - y_t$ (unit time step).

2

Curvature (2-order)

The inverse radius of the circle defined by three consecutive vectors (Menger curvature).



One chain of thought = one curve in representation space

Position is **where** the model is; velocity and curvature are **how it moves** — the quantities a graph cannot see.

How do we test this?

If logic controls the flow, then flows with the **same logical structure** should:

1ST ORDER

Similar velocities

How fast and in which direction the representation moves between steps.



Logic A · EN · math

2ND ORDER

Similar curvatures

How sharply the trajectory bends — the shape of the inference.



Logic A · ZH · weather



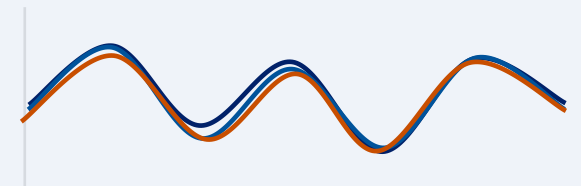
Logic A · DE · finance

INVARIANCE

Carriers shift, shape holds

Topic and language may translate a flow, but must not change its velocity / curvature pattern.

CURVATURE PROFILE $\kappa(t)$



three carriers · one profile (schematic)

EXPERIMENTAL HYPOTHESIS

Reasoning sequences sharing the **same natural-deduction skeleton** — even across different topics or languages — should exhibit **highly correlated curvature**, revealing logic as the invariant geometric structure.

04

Experiments

A dataset built to pull logic apart from its semantic carrier, and three orders of geometry measured across six models.

Dataset · One concrete case · Metrics · Findings 1–5

Two things we have to separate: logic and its carrier

① FORMAL LOGIC — the skeleton

Which steps exist, and which earlier steps each one uses.

[1] A → B

[5] A

[6] B from [1], [5]

→E

A **natural-deduction** derivation: 8–16 numbered steps, each either a premise or discharged by a rule from named earlier steps.

② SEMANTIC CARRIER — what fills the slots

The topic and the language — everything a reader sees.

weather · EN	[1] If moisture converges, then thunderclouds develop.
weather · ZH	[1] 如果水汽辐合，那么就会形成雷暴云。
finance · EN	[1] If coverage exceeds 3.0x, then interest is servable.

Step [6] still reads “from [1],[5]” in every one of them.

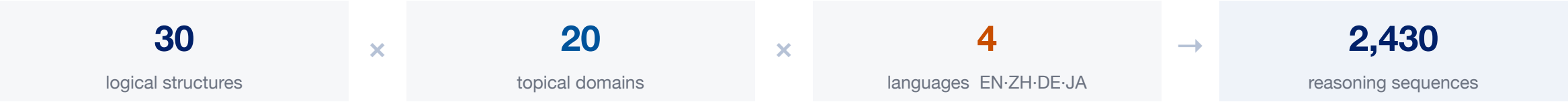
③ WHY BUILD IT THIS WAY

In ordinary text the two are **confounded**: two proofs about weather share a topic *and* usually a proof shape, so a similarity tells you nothing about which one caused it.



SO WE CROSS THEM DELIBERATELY

Hold the skeleton fixed and vary the carrier; hold the carrier fixed and vary the skeleton. Then a similarity **can only** be attributed to the factor that was held fixed.



Generated in two stages with GPT-5: an abstract skeleton first, then a rewrite that may not change the step count, the indices, or the dependencies.

One datapoint, end to end: the same 9 steps in two languages

SKELETON · LOGIC A	TOPIC: WEATHER · EN	TOPIC: WEATHER · ZH
[1] $A \rightarrow B$	If moisture converges over the city, then thunderclouds develop.	如果水汽在城市上空辐合，那么就会形成雷暴云。
[2] $B \rightarrow C$	If thunderclouds develop, then heavy rain occurs.	如果形成雷暴云，那么就会出现强降雨。
[3] $\forall x (H(x) \rightarrow J(x))$	For any location x, if a cold front passes x, then temperatures drop at x.	对任意地点 x，如果冷锋经过 x，那么 x 处气温下降。
[4] $H(a)$	A cold front is passing the airport.	一股冷锋正在经过机场。
[5] A	Moisture is converging over the city.	水汽正在城市上空辐合。
[6] B from [1], [5]	From [1] and [5], thunderclouds develop.	由 [1] 和 [5] 可得，形成雷暴云。
[7] C from [2], [6]	From [2] and [6], heavy rain occurs.	由 [2] 和 [6] 可得，出现强降雨。
[8] $J(a)$ from [3], [4]	From [3] and [4], temperatures drop at the airport.	由 [3] 和 [4] 可得，机场气温下降。
[9] $C \wedge J(a)$ from [7], [8]	From [7] and [8], heavy rain occurs and temperatures drop at the airport.	由 [7] 和 [8] 可得：出现强降雨，且机场气温下降。
Identical step count, identical indices, identical dependency graph — only the carrier changes . Any similarity shared across carriers can only come from the logic .		

Skeleton and English steps: Table 2 of the paper (which prints EN/DE); the Chinese column renders the same scaffold.

Two-step data generation with GPT-5

STEP 1 · ABSTRACT LOGIC SKELETON

Prompt for Logic Pattern Generation

You are a formal logic pattern generator.

Goal: Create an abstract, domain-agnostic reasoning sequence of exactly N steps, written in symbolic form, using standard propositional/first-order logic notation.

Strict output format:

- Exactly N lines, each line starts with a bracketed index and a single formula or conclusion, e.g.:

```
[1] A -> B
[2] B -> C
[3] C -> D
[4] (D & E) -> F
[5] forall x (H(x) -> J(x))
[6] A
[7] E
[8] H(a)
[9] D (from [1-3] and [6])
[10] F & J(a) (from [4], [7], [5], [8], [9])
```

- Use only symbols from: $\neg, \wedge, \vee, \rightarrow, \leftrightarrow, \forall, \exists$, parentheses, predicate letters with uppercase (A,B,C,...) and predicate symbols like $H(x), J(x)$.
- You may include brief justifications at the end of lines in parentheses referencing earlier step indices (e.g., (from [2] and [5])).
- The sequence must be internally coherent (later steps can be derived from earlier ones), but no proof of a fixed target is required.
- No extra commentary before or after the lines. No natural-language sentences.

Parameters (provided by caller):

- N: number of steps to output.
- logic: a label for this abstract logic (optional).

$N = \{N\}$

logic = {logic}

Now produce exactly N lines.

STEP 2 · TOPIC & LANGUAGE REWRITE

Prompt for Reasoning Rewriter

You are a reasoning rewriter.

Task: Given an abstract N-step reasoning scaffold (formal symbolic lines) and a target topic, rewrite the scaffold into a topic-specific natural-language reasoning sequence with exactly the same number of steps and the same dependency structure.

Inputs (provided by caller):

- Topic: the target domain (e.g., weather, software).
- Abstract Steps (1..N): the neutral scaffold, numbered 1..N.
- N: the total number of steps.

Output requirements:

- Produce exactly N steps, each line begins with the same bracketed index as the abstract: [1] ... to [N]
- Keep step count and ordering identical to the abstract. Do not merge, split, add, or remove steps.
- Preserve the logical dependencies: if abstract step k enables k+1, your rewrite must preserve that relationship in the topic.
- Use concrete domain terms appropriate to the topic, but keep sentences concise and precise.
- No extra commentary before or after the steps.

Multilingual mode (when Languages : are specified by the caller):

- Create a separate section for each requested language code.
- Each section starts with a header line === <code> === (e.g., === en ===).
- Under each header, write the N steps with bracketed indices [1] ... [N] in that language.
- Keep the content aligned across languages (same meaning per step index).

Inputs you will receive:

Topic: {topic}

Abstract Steps (1..N): {ABSTRACT_STEPS}

N = {N}

Now perform the rewrite.

Step count, indexing and dependency structure are held **identical** across all instantiations.

Experimental setup

Models

- Qwen1.5 0.5B, Qwen2 0.5B
- Qwen3 0.6B / 1.7B / 4B
- LLaMA3 8B
 - Two scaling axes: model size and model family.

Pipeline

- Extract hidden states for each reasoning prefix.
- **Mean-pool per step** → one vector y_t per sentence.
- Stack into a reasoning flow $Y = [y_1, \dots, y_T]$.

What we compare

- **Positions** (0-order), **velocities** (Δy , 1-order), **curvature** (Menger, 2-order).
- Grouped by three criteria: **logic**, **topic**, **language**.

Measures and baseline

- Mean **cosine similarity** for position and velocity.
- Pearson **correlation** for curvature.
- Baseline: **random shuffle** of the order of logical inferences.

Findings 1–3: logic emerges at higher order

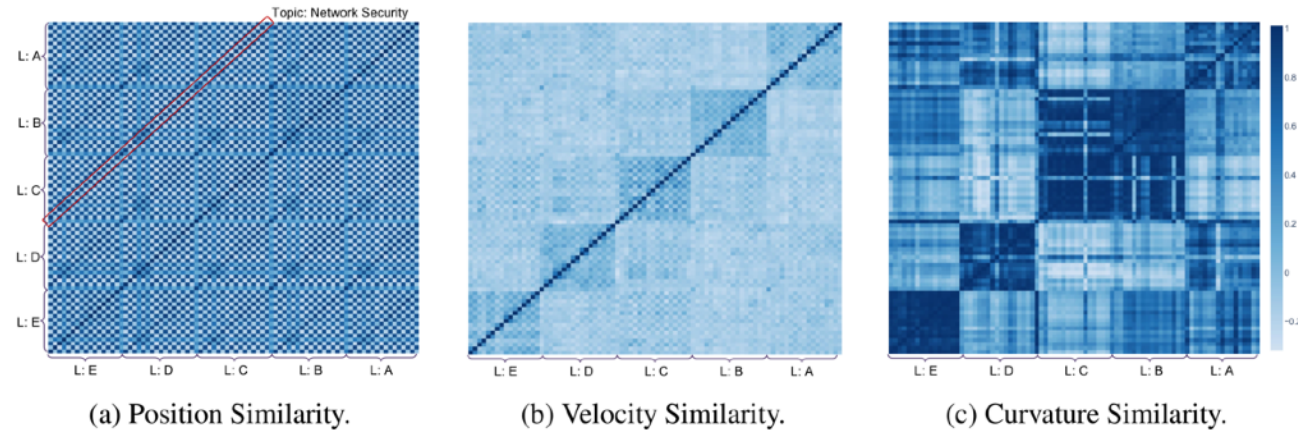


Figure 2: reasoning-flow similarity on Qwen3 0.6B. Blocks are logic templates L:A–E, instantiated across topics and languages.

FINDING 1

Positions are dominated by semantics

Same topic or language clusters tightly — the bright diagonal blocks in (a).

FINDING 2

Velocities are dominated by logic

Velocity similarity: logic $\gg$ topic / language.
The skeleton overrides the carrier in (b).

FINDING 3

Curvature sharply reveals logic

Curvature correlation aligns by logic; logics B and C even cluster together in (c).

LLMs internalise logic as **flow dynamics**, not surface-form statistics.

Findings 4–5: order matters, scale does not

Finding 4 — Random shuffle

- Randomly permute the order of reasoning steps.
- Velocity ↓ and curvature ↓ **collapse** (0.02–0.04); position stays **high** (0.30 / 0.33 / 0.81).



→ Same points, no flow: order is what the higher-order geometry sees.

Finding 5 — Stable scaling effect

- Two axes: **model size** (0.6B → 1.7B → 4B) and **model family** (Qwen1.5 / 2 / 3, LLaMA3).
- Patterns stay **stable across scale and architecture** — scaling does not alter the geometry.

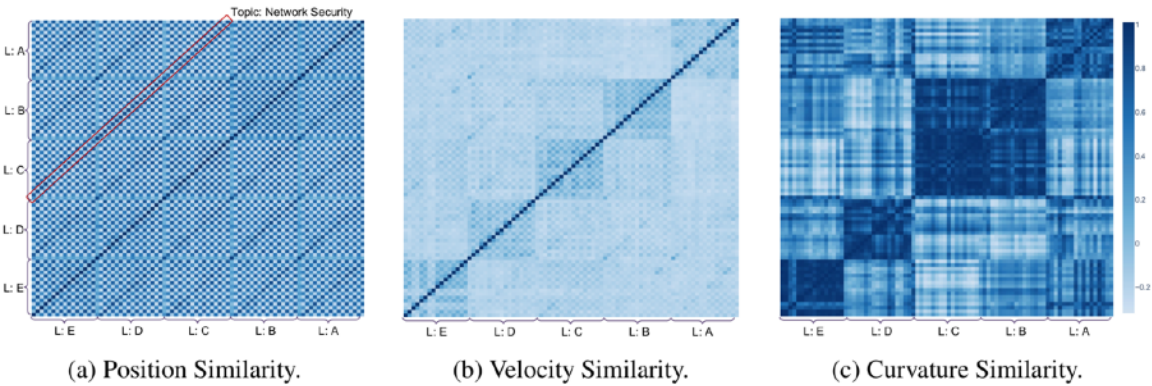


Figure 5: Llama3 8B reproduces the same block structure.

Model	Position Similarity			Velocity Similarity			Curvature Similarity		
	Logic	Topic	Lang.	Logic	Topic	Lang.	Logic	Topic	Lang.
Qwen1.5 0.5B	0.26	0.30	0.80	0.17	0.07	0.08	0.52	0.11	0.15
Qwen2 0.5B	0.21	0.24	0.85	0.15	0.06	0.07	0.52	0.10	0.12
Qwen3 0.6B	0.26	0.30	0.85	0.17	0.07	0.08	0.53	0.11	0.13
Qwen3 1.7B	0.44	0.46	0.89	0.19	0.08	0.09	0.46	0.13	0.15
Qwen3 4B	0.33	0.35	0.86	0.16	0.07	0.08	0.53	0.11	0.13
LLaMA3 8B	0.31	0.34	0.74	0.15	0.06	0.07	0.58	0.13	0.17
Random Shuffle	0.30	0.33	0.81	0.02	0.02	0.04	0.02	0.03	0.04

Table 1: the same pattern across six models.

05

Significance

What a geometric account of reasoning buys us — against the stochastic-parrot reading, across symbolic and connectionist traditions, and where the research goes next.

Stochastic parrots · Unification · Platonic vs Aristotelian · Future work

Challenging the “stochastic parrots” view

THE THESIS

Stochastic parrots

- LLMs generate text by statistical pattern matching.
- No genuine understanding — only surface-level co-occurrence modelling.
- Meaning is not internally represented, only mimicked.

OUR EVIDENCE AGAINST A PURELY SURFACE ACCOUNT

Carrier invariance

Logical structure persists across topic and language variations.

Higher-order geometry

Velocity and curvature capture stable reasoning dynamics.

Order sensitivity

Shuffling collapses higher-order structure while preserving surface similarity.

Model-agnostic stability

Patterns persist across scale, architecture and training data.

If models were merely parroting statistics, these invariants would **vanish** under surface perturbation. Instead we observe stable geometric constraints.

Significance

Conceptual

- Reasoning = **flow on a concept manifold**.
- Logic = **controller** regulating the differential geometry of that flow.
- Language representation = topic (0-order) + language (0-order) + **logic (1-order)**.

For interpretability

- We can **measure** reasoning trajectories — velocity and curvature.
- Tools to analyse the **continuity of a thought process**, not just its endpoints.

Unifying three traditions

- **Connectionist:** continuous embedding manifold — neural dynamics.
- **Conceptual:** geometry of regions, topology, similarity structure.
- **Symbolic:** logical constraints as invariants on the manifold.

The three accounts are not rivals: they describe the same object at different orders of derivative.

Toward a general law of reasoning

Aligned with the Platonic Representation Hypothesis

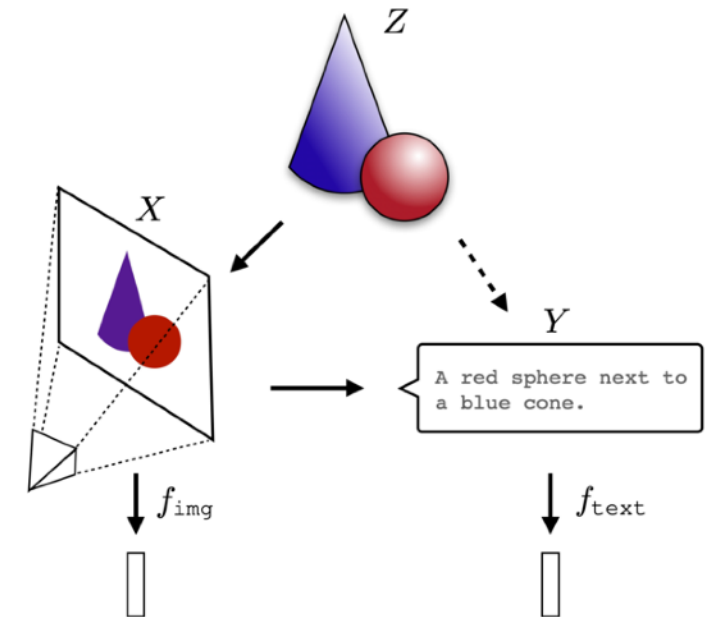
- Diverse architectures and datasets converge to **shared latent representations**.
- Logical structure emerges because efficient compression requires internalising the underlying logic.

A post-hoc, model-agnostic principle

- Suggests a **post-hoc, model- and training-agnostic principle** governing reasoning dynamics.
- Evidence for deeper, potentially **universal constraints** linking machine representations and human linguistic meaning.

The Platonic Representation Hypothesis

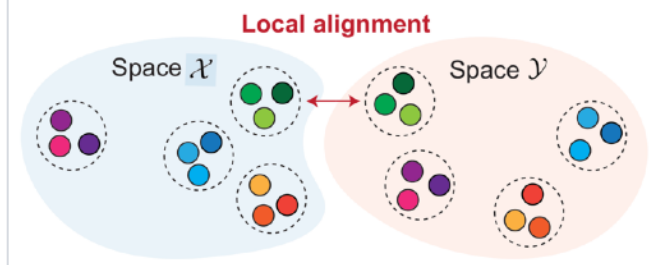
Neural networks, trained with different objectives on different data and modalities, are converging to a shared statistical model of reality in their representation spaces.



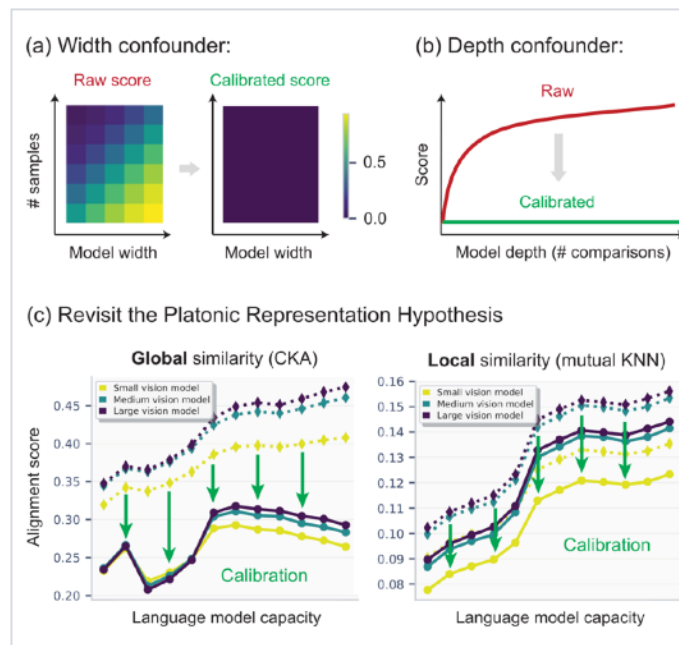
A correction to that hypothesis

The Aristotelian Representation Hypothesis

Neural networks, trained with different objectives on different data and modalities, are converging to shared *local neighborhood relationships*.



- Similarity metrics are **confounded by network scale** — depth and width **inflate** the score.
- Their fix: a **permutation null calibration** with statistical guarantees.



HOW IT RELATES TO US

Convergence, where it survives, is **local**. Velocity and curvature are local differential quantities — the same level of description.

WHAT IT SUGGESTS

And a general caution for this literature: a raw similarity score can be driven by **model scale** rather than by shared structure.

After calibration, **global** spectral convergence largely **disappears**; what survives is **local neighbourhood** similarity.

Gröger, F., Wen, S., & Brbić, M. (2026). Revisiting the Platonic Representation Hypothesis: An Aristotelian View. ICML 2026 (arXiv:2602.14486).

Future work: a catalogue of representational regularity

Logic is one regularity. The next step is to find the others, bound how far they reach, and explain why they are there.

What else is invariant?

- Beyond natural deduction: **proof strategy** (induction, contradiction, case split), **arithmetic carry**, **discourse structure**, **dialogue moves**, **code control flow**.
- Build carrier-invariant datasets for each, in the same way — vary the carrier, hold the structure fixed.
- Deliverable: a **map of which linguistic structures show up at which order** of the representation geometry.

How large is the effect?

- Measure against a **calibrated null**, not raw similarity (cf. Gröger et al., 2026).
- Vary **depth, width, family, training data, language**: where does the invariant weaken, and where does it break?
- Deliverable: **effect sizes with statistical guarantees**, not just visible block structure.

Why is it there?

- Karkada et al. (2026) trace word-level manifolds to a **translation symmetry of co-occurrence statistics**. What is the analogous symmetry one level up?
- Candidate: logical form is what stays fixed when the carrier varies, so **compression** forces it into the derivatives.
- Deliverable: a **derivation**, tested by perturbing the training distribution.

And then the payoff: **better alignment and understanding** — if we can name the regularities a model relies on, we can **measure** them, **steer at the level of logical transitions** rather than token outputs, and say concretely what a model has internalised.

Takeaways

Reasoning is a flow, not a walk.

Chain-of-thought traces a smooth, directed trajectory on the representation manifold.

Logic lives in the derivatives.

Position encodes topic and language; velocity and curvature encode the logical skeleton.

The invariant is carrier-free.

Across 30 skeletons, 20 topics and 4 languages, curvature correlates by logic — not by surface form.

The structure is robust.

It survives model scale and architecture, and collapses only when the logical order itself is destroyed.

Thank you

Questions and discussion

The Geometry of Reasoning: Flowing Logics in Representation Space

Yufa Zhou, Yixiao Wang, Xunjian Yin, Shuyan Zhou, Anru R. Zhang · Duke University · ICLR 2026

